

Percentage of Interview Questions	Topic	Things to Cover	Graduation Criteria (Entry level Data Scientist)	Estimated Time to Prepare from Scratch
Foundational Skills				
60%	Programming Fundamentals - Python	• Create scripts using functions, loops, conditionals • Work with data structures (lists, dictionaries, sets) • Understand basic OOP concepts • Write clean, documented code • Solve basic algorithmic problems	You can do Leetcode 10 Easy Python Questions	2 Months
	SQL & Databases	Write complex queries with joins, aggregations • Perform database CRUD operations • Optimize queries • Design simple database schemas	You can do SQL 50 on Leetcode. Try to do it with Pandas and PySpark as well	1 Month
20%	Statistics & Probability	• Apply hypothesis testing • Understand p-values and significance • Use statistical models (ANOVA, Chi-squared) • Interpret statistical results • Apply correlation analysis	You understand most concepts in the First Half of 'Practical Statistics for Data Scientists'	1 Month
	Conventional ML	• Implement and evaluate key algorithms (regression, tree-based, clustering) on real datasets • Build end-to-end ML pipelines with proper preprocessing, validation, and hyperparameter tuning • Select appropriate evaluation metrics and interpret model outputs/feature importance • Deploy a production-ready ML model that solves a meaningful problem	You understand most concepts in the Second Half of 'Practical Statistics for Data Scientists'	1 Month
10%	Deep Learning	• Build and train neural networks using a major framework (TensorFlow/PyTorch, I would recommend PyTorch) • Implement CNNs for vision tasks and RNNs/Transformers for sequential data • Apply transfer learning and optimization techniques to improve model performance • Complete one end-to-end project demonstrating practical application of deep learning	You are familiar and can implement most concepts covered in this Patrick Loeber's course	1 Month
	GenAI/LLMs	• Implement effective prompt engineering techniques for different LLM applications • Create a retrieval-augmented generation (RAG) system connecting LLMs with external data • Build a simple Agentic Framework with LangGraph • Build a UI around it using StreamLit	• Implement 2 Tutorials on LangChain website to use RAG • Implement 1 Tutorial on LangGraph website to build Agentic Framework	1 Month
Specialized AI Areas				
5%	Recommendation Engines	• Implement collaborative filtering and content-based recommendation algorithms • Build and evaluate a hybrid recommendation system on a real-world dataset	- Diego Maniloff's talk in PyCon 2015 - Implement a recommendation engine project using SKLearn/Pandas - Bonus- if you can implement it in PySpark as well	1 week
	Time Series Analysis	• Apply forecasting methods (ARIMA, Prophet, LSTM) to predict future values • Process and transform time series data to handle seasonality, trends, and anomalies	- Aileen Nielsen's talk in PyCon 2017 - Implement a time series project using SKLearn/Pandas	1 week
	Natural Language Processing	• Implement text preprocessing, vectorization, and embeddings for NLP tasks • Build models for key NLP applications (sentiment analysis, classification, named entity recognition)	- Kevin Mrkham's talk in PyCon 2016 - Implement an NLP project using SKLearn/Pandas - Bonus- if you can use spacey library for that	2 weeks
Other Skills				
5%	MLOps	• Design and implement CI/CD pipelines for machine learning models • Monitor model performance and implement automated retraining strategies	Can understand most of the concepts in this Krish Naik's video	1 week
	Cloud	- Familiarity with basic services of any popular cloud provider, preferably AWS	Can understand most of the concepts in this Suleyman's video	1 week
	Data Visualization	• Create insightful static and interactive visualizations using libraries like matplotlib, seaborn, and Plotly • Design effective dashboards that communicate key insights to technical and non-technical audiences	Can understand most of the concepts in this Alex's video	1 week
	Git	• Manage code using Git workflows including branching, merging, and resolving conflicts • Collaborate effectively on team projects using Git and GitHub/GitLab	Can understand most of the concepts in this short video	2 hours
	Virtual Environments	• Set up and manage isolated Python environments using tools like venv, conda, or poetry • Create reproducible project environments with dependency management	Can understand most of the concepts in this short video	2 hours
	API Building	• Develop RESTful APIs using FastAPI or Flask to serve machine learning models • Implement proper request validation, error handling, and documentation	Can understand most of the concepts in this short video	1 day